LECTURE NOTES ON NUMERICAL ANALYSIS

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ABSTRACT. The following material made up a portion of the second half of the Honors Applied Numerical Methods (151BH) at UCLA from the Spring 2023 quarter. Much of it is borrowed directly from Trefethen and Bau [1]. Credit is owed to Nicholas Hu, who served as the class teaching assistant and contributed significantly to the course development.

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1. CONDITIONING AND STABILITY

1.1. **General aspects.** Loosely speaking, a *well-conditioned* problem is one with the property that all small perturbations to the problem input data x lead to correspondingly small changes in the problem output $f(x)$. In contrast, for an *ill-conditioned* problem a small change in the problem input x leads to a large change in the output $f(x)$.

Let δx denote a small input perturbation, and let $\delta f := f(x+\delta x) - f(x)$ be the corresponding change in output. We define the absolute and relative condition numbers of a problem as follows. Since we typically are working on problems with vector space structures, we assume we have a norm $\|\cdot\|$ defined on both input and output spaces of f.

Definition 1.1. *The absolute condition number of a problem is defined as*

$$\widehat{\kappa} = \lim_{\delta \downarrow 0} \sup_{\|\delta x\| \leq \delta} \frac{\|\delta f\|}{\|\delta x\|}$$

and the relative condition number of problem is

$$\kappa = \lim_{\delta \downarrow 0} \sup_{\|\delta x\| \leq \delta} \frac{\left(\|\delta f\| / \|f(x)\|\right)}{\left(\|\delta x\| / \|x\|\right)} \tag{1}$$

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Both notions are important, but the relative condition number is particularly relevant for scientific computing since floating point arithmetic introduces errors that are constant in the relative sense. Generally speaking a well conditioned problem has a small κ (e.g. 1, 10, 10^2) while an ill conditioned problem has a large κ (e.g. 10^8, 10^{16}).

We note that if a problem ouput f is Fréchet differentiable with respect to the inputs, then by definition we have

$$\delta f = f(x+\delta x) - f(x) = D_f(x)\delta x + o(\delta x)$$

where $D_f(x)$ is the Fréchet derivative of f. With a little work, we can show that $\hat{\kappa}$ equals the norm of $D_f(x)$.

Theorem 1.2. *Suppose X and Y are two normed linear spaces and that $f : X \to Y$ is Fréchet differentiable at $x \in X$. Then*

$$\widehat{\kappa} = \|D_f(x)\|$$

where $\|D_f(x)\| = \sup_{\|u\|_X=1} \|D_f(x)u\|_Y$ is the operator norm.

Proof. Then by definition, for any $\epsilon > 0$ there exists $\delta = \delta(\epsilon) > 0$ such that

$$\frac{\|f(x+h) - f(x) - D_f(x)h\|_Y}{\|h\|_X} < \epsilon \tag{2}$$

for any nonzero $h \in X$ such that $\|h\|_X \leq \delta$. Fix $\epsilon > 0$, and let $h \neq 0$ satisfy $\|h\|_X \leq \delta = \delta(\epsilon)$. Then

$$\begin{aligned}
\|f(x+h) - f(x)\|_Y &= \|f(x+h) - f(x) - D_f(x)h + D_f(x)h\|_Y \\
&\leq \|f(x+h) - f(x) - D_f(x)h\|_Y + \|D_f(x)h\|_Y \\
&< \epsilon\|h\|_X + \|D_f(x)h\|_Y \\
&\leq (\epsilon + \|D_f(x)\|)\|h\|_X.
\end{aligned}$$

Rearranged, this gives

$$\frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} < \epsilon + \|D_f(x)\| \tag{3}$$

and hence

$$\sup_{0<\|h\|_X\leq\delta} \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} < \epsilon + \|D_f(x)\|. \tag{4}$$

Next, consider that by definition of the supremum, $\forall \epsilon > 0$ there exists some $\tilde{u}$ with unit norm such that

$$\sup_{\|u\|=1} \|D_f(x)u\|_Y = \|D_f(x)\| < \|D_f(x)\tilde{u}\|_Y + \epsilon$$

so of course we have

$$-\epsilon + \|D_f(x)\| < \|D_f(x)\tilde{u}\|_Y.$$

Now take ϵ and δ as in (4). Multiplying both sides of the inequality by δ gives

$$-\delta\epsilon + \delta\|D_f(x)\| < \delta\|D_f(x)\tilde{u}\|_Y = \|D_f(x)\tilde{h}\|_Y.$$

where we set $\tilde{h} = \delta\tilde{u}$. By the previous inequality and the triangle inequality, we have

$$\begin{aligned}
-\delta\epsilon + \delta\|D_f(x)\| < \|D_f(x)\tilde{h}\|_Y &\leq \|D_f(x)\tilde{h} - (f(x+\tilde{h}) - f(x))\|_Y + \|f(x+\tilde{h}) - f(x)\|_Y \\
&< \epsilon\|\tilde{h}\|_X + \|f(x+\tilde{h}) - f(x)\|_Y,
\end{aligned}$$

where we also used (2) (since $\|\tilde{h}\|_X \leq \delta$). Rearranging this inequality then gives

$$\frac{\|f(x+\tilde{h}) - f(x)\|_Y}{\|\tilde{h}\|_X} > \frac{\delta\|D_f(x)\|}{\|\tilde{h}\|_X} - \epsilon - \frac{\delta\epsilon}{\|\tilde{h}\|_X} = \|D_f(x)\| - 2\epsilon$$

since $\|\tilde{h}\|_X = \delta$. Since we of course have

$$\frac{\|f(x+\tilde{h}) - f(x)\|_Y}{\|\tilde{h}\|_X} \leq \sup_{0<\|h\|_X\leq\delta} \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X},$$

we can combine what we've just shown with (4) to finally give that

$$-2\epsilon + \|D_f(x)\| < \sup_{0<\|h\|_X\leq\delta} \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} < \epsilon + \|D_f(x)\|.$$

Taking the limit as $\delta \downarrow 0$ on each side of the inequality then gives

$$-2\epsilon + \|D_f(x)\| < \hat{\kappa} < \epsilon + \|D_f(x)\|.$$

Since ϵ can be arbitrarily small, we conclude that $\hat{\kappa} = \|D_f(x)\|$ as desired. □

By a similar proof, we can show that for differential problems the relative condition number satisfies

$$\kappa = \frac{\|D_f(x)\|}{\|f(x)\|/\|x\|}$$

Next consider a few examples and exercises.

Example 1.3. *Consider the problem of computing $f(x) = \sqrt{x}$ for $x > 0$. Since f is C^1, we have the absolute condition number*

$$\widehat{\kappa} = \frac{1}{2}\frac{1}{|\sqrt{x}|}$$

and the relative condition number

$$\kappa = \frac{1}{2}\frac{|x|^{-1/2}}{|x|^{1/2}/|x|} = \frac{1}{2}.$$

So in the relative sense, this problem is well conditioned, while in the absolute sense it becomes ill conditioned for x sufficiently small.

Exercise 1.4. *Consider the problem of computing $f(x_1, x_2) = x_1 - x_2$ from $(x_1, x_2) \in \mathbb{R}^2$. Using the L^∞ norm, compute the relative condition number κ. How does it behave as $|x_1 - x_2|$ gets small?*

Example 1.5. *Finding the roots of a polynomial is an ill conditioned problem. Consider $x^2 + 2x + 1 = (x-1)^2$ which has a double root at $x = 1$. The perturbed polynomial*

$$x^2 - 2x + 0.9999 = (x - 0.99)(x - 1.01)$$

so that a change in the roots (the output) are larger than the change in the polynomial coefficient (the input). We note that since the roots can change in proportion to the square root of change in the coefficients, the Jacobian is singular (i.e. the problem is not differentiable).

1.2. **Conditioning of linear systems of equations.** We begin with a defintion:

Definition 1.6. *The condition number of a matrix A is defined to be*

$$\kappa(A) := \|A\|\,\|A^{-1}\|.$$

Here there is an abuse of notation, since κ was used in the definition of the relative condition number of a problem above. There is also a slight abuse of language; the term "condition number" is attached to a matrix, not a problem.

If A is singular, so that A^{-1} does not exist, we say $\kappa(A) = \infty$. In the above defintion, the norm is left unspecified. Typically we are interested in the particular cases of the condition number with respect to the l^2 and l^∞ norms. In this case we denote $\kappa_2(A) = \|A\|_2\|A^{-1}\|_2$ and similarly for $\kappa_\infty(A)$.

Next we consider the condition number of the problem of solving a linear system of equations $Ax = b$. In particular we consider the case when the vector b is fixed, but the matrix A is perturbed. We will see that the relative condition number of this problem is given by the condition number of the matrix A that was just defined.

If $x \in \mathbb{R}^n$ solves the linear system $Ax = b$ and $A \in \mathbb{R}^{n\times n}$ is perturbed by an infinitesimal δA, then $x + \delta x$ will solve the new system

$$\begin{aligned} &(A + \delta A)(x + \delta x) = b \\ \Longrightarrow \quad &(\delta A)x + A(\delta x) + (\delta A)(\delta x) = 0 \end{aligned} \tag{5}$$

where we used that $Ax = b$. Intuitively we can drop the $(\delta A)(\delta x)$ term, since they are two infinitesimals multiplying each other.[1]

Next we use the definition of the relative condition number of the problem; in this case the problem input is the matrix A, and the problem output is the solution vector x. So, the ratio

$$\frac{\left(\|\delta f\|/\|f(x)\|\right)}{\left(\|\delta x\|/\|x\|\right)}$$

in (1) becomes

$$\frac{\left(\|\delta x\|/\|x\|\right)}{\left(\|\delta A\|/\|A\|\right)} = \frac{\|\delta x\|\,\|A\|}{\|x\|\,\|\delta A\|}.$$

From (5) we know that

$$\delta x = -A^{-1}(\delta A)x + \mathcal{O}((\delta A)^2)$$

[1]One can make this rigorous in the argument below using a Neumann series expansion.

so that

$$\frac{\|\delta x\| \, \|A\|}{\|x\| \, \|\delta A\|} \leq \|A\| \|A^{-1}\| + \mathcal{O}\left(\|\delta A\|\right) = \kappa(A) + \mathcal{O}\left(\|\delta A\|\right),$$

and hence

$$\kappa = \lim_{\delta \downarrow 0} \sup_{\|\delta A\| \leq \delta} \frac{\|\delta x\| \, \|A\|}{\|x\| \, \|\delta A\|} \leq \lim_{\delta \downarrow 0} \sup_{\|\delta A\| \leq \delta} \left(\kappa(A) + \mathcal{O}\left(\|\delta A\|\right)\right) = \kappa(A).$$

It can be shown that for any A and b and norm $\|\cdot\|$ there exists perturbations δA such that

$$\|A^{-1}(\delta A)x\| = \|A^{-1}\| \|\delta A\| \|x\|;$$

see Lecture 12 and Exercise 3.6 in [1] for an outline of the proof. We then have the following theorem.

Theorem 1.7. *Let b be fixed and consider the problem of computing $x = A^{-1}b$, where $A \in \mathbb{R}^{n \times n}$ and $\det(A) \neq 0$. The relative condition number of this problem with respect to perturbations in A is*

$$\kappa = \|A\| \|A^{-1}\| = \kappa(A).$$

2. Discrete least squares problems

The general problem is to solve overdetermined linear systems of equations (more equations than there are unknowns) in the following sense. Let $m > n$; given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, find $x \in \mathbb{R}^n$ such that the residual

$$r := b - Ax \in \mathbb{R}^m$$

is minimized. Typically we choose to minimize r in the sense of the l^2 norm, i.e. we want to minimize

$$\|b - Ax\|_2^2.$$

This is called a *least-squares problem.*

Geometrically, the problem has an intuitive solution; the minimizer x has the property that Ax is the orthogonal projection of b onto the range of A.

Theorem 2.1. *Suppose $A \in \mathbb{R}^{m \times n}$ (with $m \geq n$) and $b \in \mathbb{R}^m$. A vector $x \in \mathbb{R}^n$ minimizes*

$$\|r\|_2 = \|b - Ax\|_2$$

if and only if r is orthogonal to the range of A; that is,

$$\langle b - Ax, Au \rangle = \left\langle A^T b - A^T A x, u \right\rangle = 0 \qquad \forall u \in \mathbb{R}^n. \tag{6}$$

Note that (6) is equivalent to

$$A^T A x = A^T b,$$

which is called the *normal* system of equations, which are uniquely solvable if and only if A has full rank. Hence, the solution x to the least-squares problem is unique if and only if A has full rank. This follows from the fact that A is full rank if and only if $A^T A$ is nonsingular.

Before proving the theorem we recall the Pythagorean theorem from linear algebra.

Lemma 2.2. *Suppose $\{x_i\}_{i=1}^n$ is a collection of mutually orthogonal vectors in $\mathbb{R}^n$. Then*

$$\|\sum_{i=1}^n x_i\|_2^2 = \sum_{i=1}^n \|x_i\|_2^2.$$

Proof. First establish the result with direct computation for $n = 2$. The general case follows from that and induction. □

Now we prove the Theorem 2.1.

Proof. Suppose we know that $r = b - Ax$ is orthogonal to the range of A, and let $z \in \text{range}(A)$. Then $z - Ax \in \text{range}(A) \implies$

$$\langle z - Ax, b - Ax \rangle = 0 \implies \|(z - Ax) - (b - Ax)\|_2^2 = \|r\|_2^2 + \|z - Ax\|_2^2$$

by the Pythagorean theorem. This means

$$\|z - b\|_2^2 \geq \|r\|_2^2.$$

The inequality is strict assuming $z \neq Ax$, giving the desired result.

Now suppose x minimizes $\|b - Ax\|_2$. Then for $1 \le k \le m$

$$\begin{aligned}0 = \frac{\partial}{\partial x_k}\left(\langle b - Ax, b - Ax\rangle\right) &= \frac{\partial}{\partial x_k}\left(\|b\|_2^2 + \|Ax\|_2^2 - 2\langle b, Ax\rangle\right) \\ &= \frac{\partial}{\partial x_k}\Big(\sum_{i=1}^m\big(\sum_{j=1}^n a_{ij}x_j\big)^2 - 2\sum_{i=1}^m b_i\big(\sum_{j=1}^n a_{ij}x_j\big)\Big) \\ &= 2\sum_{i=1}^m a_{ik}\big(\sum_{j=1}^n a_{ij}x_j\big) - 2\sum_{i=1}^m a_{ik}b_i\end{aligned}$$

$\implies$

$$\sum_{i=1}^m a_{ik}\big(\sum_{j=1}^n a_{ij}x_j\big) = \sum_{i=1}^m a_{ik}b_i$$

which is the component form of the matrix equation $A^TAx = A^Tb$, which is equivalent to having $(b - Ax) \perp \mathrm{range}(A)$.

□

Theorem 2.1 says that whenever $A \in \mathbb{R}^{m\times n}$ has full rank, the least-squares problem has a unique solution given by

$$x = (A^TA)^{-1}A^Tb.$$

We call the matrix

$$A^\dagger := (A^TA)^{-1}A^T \in \mathbb{R}^{n\times m}$$

the *pseudo-inverse* of A.

In summary, when $A \in \mathbb{R}^{m\times n}$ has full rank, the least-squares problem problem numerically comes down to computing either $x = A^\dagger b$ and/or $y = Ax = AA^\dagger b$.

In practice, computing $x = A^+b$ is typically done in one of three ways:

i) the Cholesky decomposition
ii) the QR factorization
iii) the SVD

which we now discuss in the order just mentioned.

3. Cholesky decomposition

3.1. **General overview.** The Cholesky decomposition is a special case of Gaussian elimination (the PLU decomposition) that can be used when the matrix $M \in \mathbb{R}^{n\times n}$ to be factorized is both symmetric and positive definite. More generally, it can work whenever A is Hermitian and positive definite, but we focus on the real case here.

Recall that A is symmetric whenever $A = A^T \iff a_{ij} = a_{ji}$ for $1 \le i, j \le n$, while A is positive definite when

$$\langle u, Au\rangle > 0 \qquad \forall u \in \mathbb{R}^n, u \ne 0.$$

If the strict inequality above is relaxed to be a greater-than or equal-to, we say A is positive semi-definite.

The Cholesky decomposition is a way of writing symmetric, positive-definite A as

$$A = LL^T$$

where L is a lower triangular matrix (and hence, L^T is upper triangular). Before discussing how the Cholesky decomposition works, we briefly comment on how it can be used to solve the normal equations to compute solutions to least-squares problems, emphasizing, however, that it has a much wider range of applicability in computational mathematics.

3.2. **Cholesky decomposition for least-squares computations.** In section 2 above we showed that solving least-squares problems amounted to solving the normal equations

$$A^TAx = A^Tb. \tag{7}$$

Assuming $A \in \mathbb{R}^{m\times n}$ is full-rank, we have that $A^TA \in \mathbb{R}^{n\times n}$ is nonsingular, symmetric, and positive definite:

$$\left\langle u, A^TAu\right\rangle = \langle Au, Au\rangle = \|Au\|_2^2 > 0 \text{ for } u \in \mathbb{R}^n, u \ne 0.$$

Hence one can use the Cholesky decomposition to solve (7).

The procedure is as follows:

1. Form the matrix A^TA and the vector $\tilde{b} := A^Tb$.
2. Compute the Cholesky factorization $A^TA = LL^T$.
3. Solve $Lw = \tilde{b}$ using forward substitution.

4. Solve $L^T x = w$ using back substitution.

Note that since $A^T A$ is symmetric, we only need to compute the entries along and below the diagonal, which can save a lot of time and memory if m and n are large.

3.3. **How the Cholesky decomposition works.** To understand how the Cholesky decomposition works, it is helpful to recall some facts about symmetric and positive definite matrices.

The following is a routine exercise in linear algebra:

Exercise 3.1. *Suppose A is symmetric positive definite. Then its eigenvalues are positive real numbers, and the eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Note that the result is true for Hermitian positive definite matrices as well. We also have:

Lemma 3.2. *Suppose $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, and suppose $X \in \mathbb{R}^{n\times l}$ has full-rank with $n \geq l$. Then*
$$X^T A X$$
is also symmetric positive definite.

Proof.
$$(X^T A X)^T = X A^T X^T = X A X^T$$
and for any $u \in \mathbb{R}^n$
$$\langle u, X^T A X u\rangle = \langle Xu, AXu\rangle > 0.$$
as desired. □

Definition 3.3. *The k-th principal submatrix of $A \in \mathbb{R}^{n\times n}$ is the $k \times k$ matrix with entries a_{ij} for $1 \leq i, j \leq k$.*

Corollary 3.4. *All of the principal submatrices of a symmetric positive definite matrix A are also symmetric positive definite matrices.*

Proof. Symmetry is clear; using the previous lemma with X to be the first k columns of the $n \times n$ identity matrix gives the result. □

Example 3.5. *As an example of the previous corollary, take $A \in \mathbb{R}^{3\times 3}$ and take*
$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$
which consists of the first 2 columns of the 3×3 identity matrix. Note that X has full rank since its columns are linearly independent. Then
$$X^T A X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$
which is the 2×2 principal submatrix of A.

It is also useful to define another class of principal submatrices.

Definition 3.6. *For $0 < k < n$, define the $k \times k$ lower-right principal submatrix of $A \in \mathbb{R}^{n\times n}$ to be the matrix with entries a_{ij} for $n - k + 1 \leq i, j \leq n$.*

We also have:

Corollary 3.7. *All of the $k \times k$ lower-right principal submatrices of a symmetric positive definite matrix A are also symmetric positive definite matrices.*

Proof. As in the proof of the previous corollary, use the previous lemma, this time with X taken to be the *last* k columns of the $n \times n$ identity matrix. □

Example 3.8. *Again take $A \in \mathbb{R}^{3\times 3}$ and take*
$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$
which consists of the last 2 *columns of the 3×3 identity matrix. Note that X has full rank since its columns are linearly independent. Then*
$$X^T A X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a_{22} & a_{23} \\ a_{23} & a_{33} \end{pmatrix}$$

We're now ready to construct the Cholesky factorization. The idea is to induct on the following process: we ultimately want to write A as the product of a lower triangular matrix times its transpose. Since the product of lower triangular matrices is also lower triangular (and similary for upper triangular matrices), if we write

$$A = L_1 B_1 L_1^T$$

where L_1 is lower triangular and B_1 is symmetric positive definite and has 1st principle submatrix equal to the identity matrix, we can continue the process and write

$$A = L_1 L_2 B_2 L_2^T L_1^T$$

where now B_2 is also symmetric positive definite but now has 2nd principle submatrix equal to the identity. We then continue until $B_n = I$.

More concretely, write

$$A = \begin{pmatrix} a_{11} & \omega^T \\ \omega & K \end{pmatrix}$$

where $a_{11} \in \mathbb{R}$, $\omega \in \mathbb{R}^{(n-1)\times 1}$ and $K \in \mathbb{R}^{(n-1)\times(n-1)}$. Then observe that A can be written as

$$\begin{pmatrix} a_{11} & \omega^T \\ \omega & K \end{pmatrix} = \begin{pmatrix} \sqrt{a_{11}} & 0 \\ \frac{\omega}{\sqrt{a_{11}}} & I \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & K - \frac{\omega\omega^T}{a_{11}} \end{pmatrix} \begin{pmatrix} \sqrt{a_{11}} & \frac{\omega^T}{\sqrt{a_{11}}} \\ 0 & I \end{pmatrix}. \tag{8}$$

Note that $\sqrt{a_{11}}$ is real since a_{11} must be positive. We now need to show that $M := K - \omega\omega^T/\sqrt{a_{11}}$ is positive definite for the process to continue (it's clear from the symmetry of A that M is symmetric).

We know for any $x \in \mathbb{R}^n$ that $\langle x, Ax\rangle > 0$. For arbitrary $\tilde{x} \in \mathbb{R}^{n-1}$, consider

$$x = \begin{pmatrix} x_1 \\ \tilde{x} \end{pmatrix}, \qquad x_1 = -\frac{\omega^T \tilde{x}}{a_{11}}$$

(here the notation $\omega^T\tilde{x}$ is equivalent to saying $\langle \omega, \tilde{x}\rangle$). Then

$$0 < \langle x, Ax\rangle = a_{11}x_1^2 + 2x_1\omega^T\tilde{x} + \langle \tilde{x}, K\tilde{x}\rangle \tag{9}$$

$$= \langle \tilde{x}, K\tilde{x}\rangle - \left\langle \tilde{x}, \frac{\omega\omega^T}{a_{11}}\tilde{x} \right\rangle \tag{10}$$

which shows indeed that M is positive definite as desired. We've thus shown that

Theorem 3.9. *Every symmetric positive definite matrix $A \in \mathbb{R}^{n\times n}$ has a unique Cholesky factorization.*

We note that there is another way to easily show that the $(n-1)\times(n-1)$ matrix $K - \omega\omega^T/\sqrt{a_{11}}$ is positive definite. Note that the factorization (8) is indeed of the form

$$A = L_1 B_1 L_1^T \iff L_1^{-1} A (L_1^T)^{-1} = B_1.$$

Since A is positive definite (and since $(A^T)^{-1} = (A^{-1})^T$) we know that B_1 is positive definite. By Corollary 3.7 we immediately get the desired result.

4. QR factorization

(reference: [1])

4.1. **Projections.** A projector or projection matrix is a square matrix P such that

$$P^2 = P.$$

These are also called idempotent maps. The key point is that if $v \in \text{range}(P)$, then $Pv = v$. Given some projector P, we immediately have that $I - P$ is also a projector; we call it the complementary projector. Given a projector P, we can separate $\mathbb{R}^n$ into a direct sum of two subspaces, namely the range of P and the range of $I - P$.

Of particular interest are orthogonal projectors; in this case $\text{range}(P) \perp \text{range}(I - P)$. We have the following characterization:

Theorem 4.1. *A projector P is orthogonal if and only if $P = P^*$.*

Proof. Suppose $P = P^*$. For $\tilde{x}, \tilde{y} \in \mathbb{C}^n$, consider $x = P\tilde{x} \in \text{range}(P)$ and $y = P\tilde{y} \in \text{range}(I - P)$. Then

$$\langle x, y\rangle = \langle P\tilde{x}, (I-P)\tilde{y}\rangle = \langle \tilde{x}, P^*(I-P)\tilde{y}\rangle = \langle \tilde{x}, (P - P^2)\tilde{y}\rangle = 0.$$

Now suppose that P is an orthogal projector. Then we have that for any $x, y \in \mathbb{C}^n$

$$\langle Px, (I-P)y\rangle = 0 \implies \langle Px, Py\rangle = \langle Px, y\rangle$$

since range(P) is orthogonal to the range of its complementary projector $I-P$. We also have that

$$\langle (I-P)x, Py\rangle = 0 \implies \langle Px, Py\rangle = \langle x, Py\rangle .$$

Combining the two equalities gives

$$\langle Px, y\rangle = \langle Px, Py\rangle = \langle x, Py\rangle$$

which shows $P = P^*$ as desired. □

Exercise 4.2. *Let $P \in \mathbb{C}^{n\times n}$ be a nonzero projector. Show that $\|P\|_2 \geq 1$ with equality if and only if P is an orthogonal projector.*

4.2. **Gram-Schmidt and modified Gram-Schmidt orthogonalization.** Suppose that $q_1 \in \mathbb{C}^n$ is a unit vector, and define $\widetilde{Q}_1 : \mathbb{C}^n \to \mathbb{C}^n$

$$\widetilde{Q}_1 := q_1 q_1^*.$$

It is easy to see (using the theorem just given) that this is an orthogonal projection onto the one-dimensional subspace spanned by q_1. Consider now the set of mutually orthonormal vectors $\{q_1, \ldots, q_k\}$; then the complementary projector

$$I - \widetilde{Q}_1$$

is an orthogonal projection onto the subspace spanned by $\{q_2, \ldots, q_k\}$.

We can also define

$$\hat{Q}_2 := \left[q_1 \middle| q_2 \right] \in \mathbb{C}^{n\times 2}$$

and

$$\widetilde{Q}_2 = \hat{Q}_2 \hat{Q}_2^* \implies \widetilde{Q}_2 v = \langle q_1, v\rangle\, q_1 + \langle q_2, v\rangle\, q_2$$

for any $v \in \mathbb{C}^n$. So, $\widetilde{Q}_2$ is the orthgonal projection onto the subspace spanned by q_1 and q_2.

More generally, we can define

$$\hat{Q}_k := \left[q_1 \middle| q_2 \middle| \ldots \middle| q_k \right] \in \mathbb{C}^{n\times k}$$

and

$$\widetilde{Q}_k = \hat{Q}_k \hat{Q}_k^* \implies \widetilde{Q}_k v = \sum_{i=1}^{k} \langle q_i, v\rangle\, q_i$$

for any $v \in \mathbb{C}^n$. These matrices are the building blocks of the Gram-Schmidt orthogonalization process.

Recall that given any set of linearly independent vectors $\{a_1, a_2, \ldots, a_k\}$ (where each $a_i \in \mathbb{C}^n$), we can transform them into an mutually orthonormal set. Starting with a_1

$$q_1 = \frac{a_1}{\|a_1\|}$$

we then say

$$\begin{aligned}
v_2 &= (I - \widetilde{Q}_1)a_2 = a_2 - \langle q_1, a_2\rangle\, q_1, \qquad q_2 = \frac{v_2}{\|v_2\|} \\
v_3 &= (I - \widetilde{Q}_2)a_3 = a_3 - \langle q_1, a_3\rangle\, q_1 - \langle q_2, a_3\rangle\, q_2, \qquad q_3 = \frac{v_3}{\|v_3\|} \\
&\vdots \\
v_k &= (I - \widetilde{Q}_{k-1})a_k = a_k - \sum_{i=1}^{k-1} \langle q_i, a_k\rangle\, q_i, \qquad q_k = \frac{v_k}{\|v_k\|}.
\end{aligned}$$

This of course can be rearranged to read

$$\begin{aligned}
a_1 &= r_{11} q_1 \\
a_2 &= r_{12} q_1 + r_{22} q_2 \\
a_3 &= r_{13} q_1 + r_{23} q_2 + r_{33} q_3 \\
&\vdots \\
a_k &= r_{1k} q_1 + r_{2k} q_2 + \ldots r_{k,k-1} q_{k-1} + r_{kk} q_k
\end{aligned}$$

where $r_{ij} := \langle q_i, a_j \rangle$ (for $i < j$) and $r_{ii} = \|a_i - \sum_{j=1}^{i-1} r_{ji} q_j\|$. Hence, we can think of each a_i for $1 \leq i \leq k$ as a linear combination of the vectors $\{q_i\}_{j=1}^{i-1}$. This is exactly the case of matrix multiplication of

$$\hat{Q}_k = \left[q_1 \middle| q_2 \middle| \dots \middle| q_k \right]$$

by upper triangular matrix $R \in \mathbb{C}^{n \times n}$. Ergo we've derived the decomposition $A = QR$.

In finite precision arithmetic, this method is problematic, as orthogonality when k is large can be compromised due to rounding error. In practice it is better to use a *modified* version of this process based on the following observation.

For simplicity, first consider the case when $k = 2$. The orthogonal projection $I - \widetilde{Q}_2$ can be computed as a sequence of two separate projections $(I - q_2 q_2^*)$ and then $(I - q_1 q_1^*)$. Indeed, for any $v \in \mathbb{C}^n$

$$\begin{aligned} (I - q_2 q_2^*)(I - q_1 q_1^*)v &= (I - q_2 q_2^*)(v - \langle q_1, v \rangle q_1) \\ &= v - \langle q_2, v \rangle q_2 - \langle q_1, v \rangle q_1 + \langle q_1, v \rangle \langle q_2, q_1 \rangle q_2 = (I - \widetilde{Q}_2)v \end{aligned}$$

since $q_1 \perp q_2$. More generally, we can easily extend this to say that

$$I - \widetilde{Q}_k = (I - q_k q_k^*)(I - q_{k-1} q_{k-1}^*) \dots (I - q_1 q_1^*)$$

which means we can rewrite the Gram-Schdmit process as

$$\begin{aligned} q_1 &= \frac{a_1}{\|a_1\|} \\ v_2 &= (I - q_1 q_1^*)a_2, \qquad q_2 = \frac{v_2}{\|v_2\|} \\ v_3 &= (I - q_2 q_2^*)(I - q_1 q_1^*)a_3, \qquad q_3 = \frac{v_3}{\|v_3\|} \\ &\vdots \\ v_k &= (I - q_{k-1} q_{k-1}^*) \dots (I - q_1 q_1^*)a_k, \qquad q_k = \frac{v_k}{\|v_k\|}. \end{aligned}$$

which we call the *modified Gram-Schmidt (GS) process.* To be clear, this is the exact same procedure as the "regular" version in exact arithmetic; however, it is more numerically stable to rounding errors.

4.3. **Forming the QR factorization.** The modified GS process can be put written as an algorithm by noting that at the second step, we need to apply the projection $I - q_1 q_1^*$ to vectors a_2 through a_k, while at the third step, we need to apply the projection $I - q_2 q_2^*$ to vectors a_3 through a_k, and so on. Each application of a projector involves a computing a scalar inner product; as before with the "regular" GS, we will store these values to form the upper triangular factor R in the decomposiion $A = QR$.

The algorithm pseudocode is given by the following. Note that the first for-loop is typically not

Algorithm 1 Modified Gram-Schmidt algorithm

procedure MGS($A \in \mathbb{C}^{n \times k}$) ▷ k linearly independent vectors
    for $i = 1$ to k **do**
        $v_i = a_i$
    for $i = 1$ to k **do**
        $r_{ii} = \|v_i\|$
        $q_i = v_i / r_{ii}$
        for $j = i + 1$ to k **do**
                $r_{ij} = \langle q_i, v_j \rangle$
                $v_j = v_j - r_{ij} q_i$

    return $Q \in \mathbb{C}^{n \times k}$, $R \in \mathbb{C}^{n \times n}$

carried out in practice, as the vectors a_i are directly overwritten by v_j.

5. The Singular Value Decomposition (SVD)

5.1. **Basic facts.** *Any* matrix $A \in \mathbb{C}^{m\times n}$ (no matter the dimensions or the rank) can be factorized as

$$A = U\Sigma V^*$$

where

$$U \in \mathbb{C}^{m\times m} \text{ is unitary,}$$
$$\Sigma \in \mathbb{C}^{m\times n} \text{ is diagonal,}$$
$$V \in \mathbb{C}^{n\times n} \text{ is unitary.}$$

The diagonal entries of Σ are called the singular values of A, and they are all real and non-negative. Typically they are ordered from largest to smallest

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_p \geq 0$$

where $p = \min\{m, n\}$. The columns of U and V are called left and right singular vectors, respectively.

Geometrically, the SVD says that the action of applying A to the unit ball in $\mathbb{R}^n$ can be broken down into first rotating it (multiplication by V^*), then stretching the rotated coordinate axes in the j-th direction by σ_j (multiplication by Σ), and then rotating the result (multiplication by U).

Theorem 5.1. *The nonzero singular values of $A \in \mathbb{C}^{m\times n}$ equal the square roots of the nonzero eigenvalues of both A^*A and AA^*.*

Proof. Since

$$A^*A = V\Sigma^*\Sigma V^* \in \mathbb{C}^{n\times n}$$

this means that the entries of $\Sigma^*\Sigma \in \mathbb{R}^{n\times n}$ are the eigenvalues of A^*A, and they are $\sigma_1^2, \sigma_2^2, \ldots, \sigma_p^2$, where $p = \min\{m, n\}$ (if $n > m$ then there are an additional $n - m$ zero eigenvalues).

A similar calculation holds for AA^*. □

We can also use the spectral decomposition of A^*A to prove existence of the SVD.

Theorem 5.2. *Any matrix $A \in \mathbb{C}^{m\times n}$ possesses a singular value decomposition $A = U\Sigma V^*$.*

Proof. Since $M := A^*A \in \mathbb{C}^{n\times n}$ is a (positive semi-definite) Hermitian operator, by the spectral theorem there exists a unitary matrix $V \in \mathbb{C}^{n\times n}$ such that

$$M = V\Lambda V^*. \tag{11}$$

Let $D = \text{diag}(\lambda_1, \ldots, \lambda_r)$ be a diagonal matrix consisting of the positive eigenvalues of M (here $r = \text{rank}(M) \leq n$). Let $\{v_i\}_{i=1}^r$ and $\{v_i\}_{i=r+1}^n$ be the eigenvectors of M corresponding to non-zero and zero eigenvalues of M, respectively, and take these eigenvectors to be mutually orthonormal (this can be done by the spectral theorem for Hermitian operators).

For $1 \leq i \leq r$, let v_i be an eigenvector corresponding to a non-zero eigenvalue. By definition,

$$A^*Av_i = \lambda_i v_i.$$

Let $\sigma_i := \sqrt{\lambda_i}$, and define

$$u_i = \frac{Av_i}{\sigma_i}. \tag{12}$$

For $1 \leq i, j \leq r$, we have

$$\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle Av_i, Av_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle A^*Av_i, v_j\rangle = \frac{\lambda_i}{\sigma_i\sigma_j}\langle v_i, v_j\rangle = \delta_{ij},$$

so that $\{u_i\}_{i=1}^r$ is a mutually orthonormal collection. If we order the singular values such that $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r \geq 0$, take $\hat{\Sigma}_r = \text{diag}(\sigma_1, \ldots, \sigma_r)$, and define

$$\hat{V}_r = \Big[v_1\Big|v_2\Big|\ldots\Big|v_r\Big], \qquad \hat{U}_r = \Big[u_1\Big|u_2\Big|\ldots\Big|u_r\Big],$$

then it follows from (12) that

$$A\hat{V}_r = \hat{U}_r\hat{\Sigma}_r. \tag{13}$$

Next, pad $\hat{V}_r$ with the remaining eigenvectors of M, so that

$$V = \Big[v_1\Big|v_2\Big|\ldots\Big|v_r\Big|v_{r+1}\Big|\ldots\Big|v_n\Big] \in \mathbb{C}^{n\times n},$$

is unitary, and hence $VV^* = V^*V = I \in \mathbb{R}^{n\times n}$. Then, pad $\hat{\Sigma}_r$ with zeros so that it has dimensions $r \times n$. We can then right-multiply both sides of the padded version of equation (13) by V^* and then drop the redundancies to arrive at

$$A = \hat{U}_r \hat{\Sigma}_r \hat{V}_r^*, \tag{14}$$

which is the "compact" SVD.

Next, we can extend the orthonormal collection $\{u_i\}_{i=1}^r$ to an othornormal basis for $\mathbb{C}^m$ (this is possible since we can always find a basis for the orthogonal complement to $\text{span}\{u_1, \ldots, u_r\}$ and then perform GS on that basis). Take U to be the matrix whose columns are the extended orthonormal basis vectors for $\mathbb{C}^m$. Then, pad $\hat{\Sigma}_r$ with zeros so that it has dimensions $m \times n$ (and call it Σ). From here we can conclude that

$$A = U\Sigma V^*.$$

□

Next, suppose that $m \geq n$ and that $A \in \mathbb{C}^{m\times n}$ is full-rank. In practice, storing the extra, redundant information in Σ (and hence in U) is not done, and instead a *reduced SVD* is computed:

$$A = \hat{U}\hat{\Sigma}V^*. \tag{15}$$

As before $V \in \mathbb{C}^{n\times n}$ is unitary and Σ is diagonal; however now $\Sigma \in \mathbb{R}^{n\times n}$ and all of the singular values $\sigma_j > 0$ by assumption. As before, $\hat{U} \in \mathbb{C}^{m\times n}$ still has orthonormal columns.

5.2. **Low-rank approximation.** Let $A \in \mathbb{C}^{m\times n}$ have rank $r \leq p = \min\{m, n\}$. The SVD can be thought of as the sum of a r different rank-1 matrices.

Theorem 5.3. *For any $A \in \mathbb{C}^{m\times n}$ we have*

$$A = \sum_{j=1}^{r} \sigma_j u_j v_j^* \tag{16}$$

where u_j and v_j are the left and right singular vectors, respectively.

Proof. In the SVD $A = U\Sigma V^*$, write

$$\Sigma = \Sigma_1 + \Sigma_2 + \ldots + \Sigma_r$$

where

$$(\Sigma_k)_{ij} = \begin{cases} \sigma_k & i = j = k \\ 0 & \text{else.} \end{cases}$$

Then

$$\begin{aligned} A = U\Sigma V^* &= U(\Sigma_1 + \ldots \Sigma_r)V^* \\ &= U\Sigma_1 V^* + \ldots + U\Sigma_r V^* \\ &= \sigma_1 u_1 v_1^* + \ldots \sigma_r u_r v_r^* \end{aligned}$$

as desired. □

The following guarantees that if the sum (16) is truncated at the k-th term, then the resulting matrix will capture as much of the "energy" of A that is possible.

Theorem 5.4. *Let $A \in \mathbb{C}^{m\times n}$, and for $0 \leq k \leq r$, define $A_k = \sum_{j=1}^k \sigma_j u_j v_j^*$. If $k = p = \min\{m, n\}$, define $\sigma_{k+1} = 0$. Then*

$$\|A_k - A\|_2 = \min_{\substack{B\in\mathbb{C}^{m\times n} \\ \text{rank}(B)\leq k}} \|B - A\|_2 = \sigma_{k+1}$$

This property means that computing the SVD is a powerful way of forming low-rank approximations to large matrices. This makes the SVD quite useful for a large variety of fields of science and engineering, for example image processing, signal processing, reduced-order modeling, designing search engines, and weather forecasting.

Proof. First, we note that

$$\|A - A_k\|_2 = \sigma_{k+1}$$

from the fact that for any matrix $M \in \mathbb{C}^{m\times n}$, $\|M\|_2$ equals the largest nonzero singular value.

Next, note that for any matrix $B \in \mathbb{C}^{m\times n}$ with $\text{rank}(B) \leq k$, $\dim(\ker(B)) \geq n - k$ by the rank-nullity theorem. Also, since the dimension of the span of the first $k + 1$ singular vectors $\{v_1, \ldots, v_{k+1}\}$ is $k + 1$, this means that

$$S := \ker(B) \cap \text{span}\{v_1, \ldots, v_{k+1}\}$$

is nontrivial.

So, let $w \in S$ be nonzero, and suppose that $\exists B \in \mathbb{C}^{m\times n}$ such that $\|A - B\|_2 < \sigma_{k+1}$. Then

$$\|Aw\|_2 = \|(A - B)w\|_2 \leq \|A - B\|_2\|w\|_2 < \sigma_{k+1}\|w\|_2.$$

Also, writing w in the span of $\{v_1, \ldots, v_{k+1}\}$ and using the decomposition of A into rank-1 matrices gives

$$\begin{aligned}\|Aw\|_2^2 = \|\sum_{i=1}^{k+1} Ac_iv_i\|_2^2 = \|\sum_{i=1}^{k+1} c_i\sigma_iu_i\|_2^2 &= \sum_{i=1}^{k+1} |c_i|^2\sigma_i^2\|u_i\|_2^2 \\ &= \sum_{i=1}^{k+1} |c_i|^2\sigma_i^2 \geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} |c_i|^2 = \sigma_{k+1}^2\|w\|_2^2\end{aligned}$$

where we used that the left and right singular vectors are each a collection of mutually orthonormal vectors.

Putting it all together gives that

$$\sigma_{k+1} \leq \|Aw\|_2 < \sigma_{k+1}\|w\|_2$$

which is a contradiction. Hence

$$A_k = \underset{\substack{B\in\mathbb{C}^{m\times n} \\ \text{rank}(B)\leq k}}{\arg\min} \; \|B - A\|_2$$

as desired. □

5.3. Least-squares problems with the SVD: full-rank and rank-deficient cases.

The (reduced) SVD can be used to solve the least-squares problem

$$\text{Compute } x^\star = \underset{x\in\mathbb{C}^n}{\arg\min}\, \|b - Ax\|_2 \tag{17}$$

where $A \in \mathbb{C}^{m\times n}$ with $m \geq n$.

First suppose that A is full-rank. Recall that the solution to (17) is given by the solution to the normal equations

$$A^*Ax = A^*b$$

which is

$$x = A^\dagger b = (A^*A)^{-1}A^*b.$$

Here we are using the complex version of the main result that was developed in section 2. Inserting the reduced SVD (15) then gives

$$A = \hat{U}\hat{\Sigma}V^* \implies A^*A = V\hat{\Sigma}^2V^* \implies A^\dagger = V\hat{\Sigma}^{-1}\hat{U}^*.$$

So, we can compute the least-squares solution as

$$x^\star = V\hat{\Sigma}^{-1}\hat{U}^*b.$$

Next, suppose that $A \in \mathbb{C}^{m\times n}$ has rank $r < n$. In this case, there are an infinite number of solutions to the least squares problem, since A^*A is singular; however, we have the following theorem.

Theorem 5.5. *The least-squares solution with minimal 2-norm is*

$$x^\star = \hat{V}_r\hat{\Sigma}_r^{-1}\hat{U}_r^*b,$$

where $A = \hat{U}_r\hat{\Sigma}_r\hat{V}_r^*$ is the compact SVD (14).

Proof. First, using the compact SVD, we have the $x^\star$ satisfies the normal equations $A^*Ax = A^*b$, hence,

$$x^\star \in \underset{x\in\mathbb{C}^n}{\arg\min}\, \|Ax - b\|_2^2.$$

To show that, amongst this set, $x^\star$ has the smallest 2-norm, we consider the full SVD

$$A = U\Sigma V^*.$$

Since U and V are both unitary, finding

$$\min_{x\in\mathbb{C}^n} \|Ax - b\|_2^2$$

is equivalent to finding

$$\min_{w\in\mathbb{C}^n} \|\Sigma w - U^*b\|_2^2,$$

where $w = V^*x$. We can directly compute the norm as

$$\|\Sigma w - U^* b\|_2^2 = \sum_{i=1}^{m} (\sigma_i w_i - \langle u_i, b\rangle)^2, \tag{18}$$

where u_i is the i-th column of U and $\{\sigma_i\}_{i=1}^m$ are the singular values of A. The sum is clearly minimized if, for $1 \leq i \leq r$ we set $w_i = \langle u_i, b\rangle / \sigma_i$. Since $\sigma_i = 0$ for $r < i \leq n$, the choice of values $\{w_i\}_{i=r+1}^n$ do not affect the (18). To minimize the norm of w, then, we should simply set them equal to zero. This is equivalent to taking

$$w = \hat{\Sigma}^{-1} \hat{U}_r^* b$$

where $\hat{\Sigma} := \text{diag}(\sigma_1^{-1}, \sigma_2^{-1}, \ldots, \sigma_r^{-1}) \in \mathbb{R}^{n \times r}$. Since we set $w = V^* x$, this means

$$x = Vw = V\hat{\Sigma}^{-1}\hat{U}_r^* b.$$

Because of the dimensions of $\hat{\Sigma}^{-1}$, however, this is equivalent to the solution from the compact SVD

$$x = V\hat{\Sigma}^{-1}\hat{U}_r^* b = \hat{V}_r \hat{\Sigma}_r^{-1} \hat{U}_r^* b,$$

as desired. □

5.4. **Computing the SVD.** In practice, computing the SVD is nontrivial and requires, at the minimum, an understanding of numerical methods for computing eigenvalues and eigenvectors. We briefly mention that, in principle, one could:

- form A^*A
- compute the eigen-decomposition $A^*A = V\Lambda V^*$
- take Σ to be the $m \times n$ nonnegative diagonal square root of Λ
- solve the system $U\Sigma = AV$ for U (using e.g. QR-factorization).

In practice, however, this turns out to be ill-conditioned. A more numerically stable approach is instead to compute an eigen-decomposition of the block matrix

$$M := \begin{bmatrix} 0 & A^* \\ A & 0 \end{bmatrix}.$$

In particular, from the SVD we know that

$$\begin{aligned} A = U\Sigma V^* &\implies AV = U\Sigma \\ A^* = V\Sigma U^* &\implies A^*U = V\Sigma, \end{aligned}$$

and hence it follows that

$$\begin{bmatrix} 0 & A^* \\ A & 0 \end{bmatrix} \begin{bmatrix} V & V \\ U & -U \end{bmatrix} = \begin{bmatrix} V & V \\ U & -U \end{bmatrix} \begin{bmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{bmatrix}$$

is an eigen-decomposition of the form $MX = \Lambda X$. We also note that the block matrix X is unitary if it is multiplied by a factor of $1/\sqrt{2}$. See [1] for a complete description of how to use this eigen-decomposition in practice.

6. The Conjugate Gradient (CG) Method

6.1. **Definition and orthogonality properties.** Here and throughout this section, suppose that $A \in \mathbb{R}^{n \times n}$ is symmetric and positive-definite (SPD). Then for $x, y \in \mathbb{R}^n$

$$\langle x, y\rangle_A := \langle x, Ay\rangle$$

defines an inner product. So, given $b \in \mathbb{R}^n$ and SPD A, define the map $\Phi : \mathbb{R}^n \to \mathbb{R}$

$$\Phi(x) = \frac{1}{2}\langle x, x\rangle_A - \langle x, b\rangle. \tag{19}$$

Proposition 6.1. *Φ has a unique minimizer $x^\star = A^{-1}b$.*

Proof. Computing

$$\nabla\Phi(x) = Ax - b$$

means that $x^\star$ is a critical point. Also the Hessian matrix

$$H_\Phi(x) = A$$

is positive-definite $\forall x \in \mathbb{R}^n$, so that Φ is strictly convex, and hence the critical point is a unique minimizer. □

One way to view CG is that it is a line search method

$$x_{k+1} = x_k + \alpha_k p_k$$

where the search directions p_k, $k = 0, 1, 2, \ldots$ are chosen to be mutually A-orthogonal.

We take an alternative viewpoint based on A-orthogonal projections onto Krylov subspaces defined by A and b.

Definition 6.2. *Given $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$, define for $k \geq 1$ the k-th Krylov subspace*

$$\mathcal{K}_k = \mathcal{K}_k(A, b) := \mathit{span}\{b, Ab, \ldots, A^{k-1}\}.$$

Note that the dimension of $\mathcal{K}_k$ is determined by the so-called "grade" of A with respect to b. In particular, it can be shown that

$$\dim\left(\mathcal{K}_k(A, b)\right) = \min\{k, \mathrm{grade}(A, b)\}$$

where $\mathrm{grade}(A, b)$ is the degree of the monic polynomial $p(z)$ of minimal degree such that $p(A)b = 0$.

Suppose that $x^\star$ solves the linear system $Ax = b$. Starting from $x_0 = 0$, we define for $k > 0$ the k-th iterate of the CG method x_k to be

$$x_k = \underset{x\in\mathcal{K}_k}{\arg\min}\, \|x^\star - x\|_A^2 \tag{20}$$

so that x_k minimizes the "energy" norm defined by $\langle\cdot,\cdot\rangle_A$ over the k-th Krylov subspace. From this perspective, it is clear that x_k is simply the A-orthogonal projection of the solution $x^\star$ onto $\mathcal{K}_k$.

Suppose that we perform Gram-Schmidt orthogonalization with $\langle\cdot,\cdot\rangle_A$ on the vectors defining $\mathcal{K}_k$ to produce $\{p_0, \ldots, p_{k-1}\}$, where $\langle p_i, p_j\rangle_A = 0$ if $i \neq j$.[2] In particular, if we do this for $k = n$, we can write the true solution as

$$x^\star = \alpha_0 p_0 + \alpha_1 p_1 + \ldots + \alpha_{n-1} p_{n-1} = \sum_{j=0}^{n-1} \alpha_j p_j \tag{21}$$

for some $\{\alpha_j\}_{j=0}^{n-1} \in \mathbb{R}^n$.[3] From the definition (20) we then immediately get that

$$x_k = \sum_{j=0}^{k-1} \alpha_j p_j. \tag{22}$$

By orthogonality, we have that in both (21) and (22):

$$\alpha_j = \frac{\langle x^\star, p_j\rangle_A}{\langle p_j, p_j\rangle_A} = \frac{\langle b, p_j\rangle}{\langle p_j, p_j\rangle_A}. \tag{23}$$

Also, we have

$$x_k = \sum_{j=0}^{k-1} \alpha_j p_j = \sum_{j=0}^{k-2} \alpha_j p_j + \alpha_{k-1} p_{k-1} = x_{k-1} + \alpha_{k-1} p_{k-1} \tag{24}$$

and hence recover the description of CG as an optimization algorithm (a line search method) for minimizing (19). We also note that from the definition (20), it immediately follows that CG is, in theory, a direct method for solving the linear system $Ax = b$, as we show now.

Proposition 6.3. *CG will converge in at most n steps, i.e. $x_n = x^\star$.*

Proof. From the Cayley-Hamilton theorem, we know that A satisfies its own characteristic equation, so that

$$c_n A^n + c_{n-1} A^{n-1} + \ldots + c_1 A + I = 0$$

for some c_n, $\ldots$, c_1. This implies that

$$-c_n A^{n-1} - c_{n-1} A^{n-2} - \ldots - c_1 I = A^{-1},$$

and hence

$$x^\star = A^{-1} b = (-c_n A^{n-1} - c_{n-1} A^{n-2} - \ldots - c_1 I) b \in \mathcal{K}_n(A, b)$$

as desired. □

[2] Here we take the convention that the search directions p_i are not normalized.

[3] If $\dim\left(\mathcal{K}_n(A, b)\right) = m < n$, then the sum in (21) will run from $j = 0, \ldots, m - 1$.

So, to perform CG, we simply need the A-conjugate vectors $\{p_0, \ldots, p_{k-1}\}$. In principle, we can simply take the vectors that define $\mathcal{K}_k$ and do Gram-Schmidt with $\langle\cdot,\cdot\rangle_A$. From Section 4, however, we know that we need *all* the previous search directions $p_0, p_1, \ldots, p_{k-1}, p_k$ to get the new one, p_{k+1}. In practice however we can do much better and *only* need p_k to get p_{k+1}.

To see why, we first note that from (24) we immediately get an update formula for the residual:

$$r_{k+1} := b - Ax_{k+1} = r_k - \alpha_k A p_k.$$

From this update formula and a proof by induction, we can show that the residuals form a basis for the Krylov subspaces. The following theorem establishes this, and, more generally, the orthogonality properties that are responsible for the efficiency of the CG method.

Theorem 6.4. *Suppose that $x_0 = 0$ (and hence $r_0 = b$) and that $p_0 = b$. If for $k = 1, 2, \ldots$ we define*

$$x_k = x_{k-1} + \alpha_{k-1} p_{k-1},$$
$$r_k = r_{k-1} + \alpha_{k-1} A p_{k-1},$$

and p_k to be the projection of r_k onto the directions A-orthogonal to $\{p_0, \ldots, p_{k-1}\}$, then

1. $r_k \in \mathcal{K}_{k+1}$
2. $r_k \perp$ *span*$\{r_0, \ldots, r_{k-1}\}$
3. $r_k \perp_A \mathcal{K}_{k-1}$
4. $p_k \perp_A$ *span*$\{p_0, \ldots, p_{k-1}\}$

and hence we simply have

$$p_k = r_k - \frac{\langle r_k, p_{k-1}\rangle_A}{\langle p_{k-1}, p_{k-1}\rangle_A} p_{k-1}. \tag{25}$$

Note that property 3 and the definition for the search directions p_k are responsible for the simple update formula (25).

Proof. We pursue a proof by (strong) induction and hence first establish a base case. Firstly we want to show that $r_1 \in \mathcal{K}_2 = \text{span}\{b, Ab\}$. We know

$$r_1 = r_0 - \alpha_0 A p_0 = b - \alpha_0 Ab \in \mathcal{K}_2$$

as desired. Next we have

$$\langle r_1, r_0\rangle = \langle b - \alpha_0 Ab, b\rangle = \|b\|_2^2 - \alpha_0 \langle Ab, b\rangle = 0$$

since from (23), $\alpha_0 = \langle b, p_0\rangle / \langle p_0, p_0\rangle_A = \|b\|_2^2/\|b\|_A^2$. Also, defining $\mathcal{K}_0 = \{0\}$ we obviously have

$$r_1 \perp_A \mathcal{K}_0.$$

Finally, since $r_0 = p_0 = b$, we also have that $p_1 \perp_A \text{span}\{p_0\}$:

$$\langle p_1, p_0\rangle_A = \left\langle r_1 - \frac{\langle r_1, p_0\rangle_A}{\langle p_0, p_0\rangle_A} p_0, p_0 \right\rangle_A = 0.$$

Having established the base case, now assume that properties 1–4 and (25) hold for $j = 1, \ldots, k$ and show they hold for $j = k+1$.

By the inductive hypothesis (IH), we clearly have

$$r_{k+1} = r_k - \alpha_k A p_k \in \mathcal{K}_{k+2},$$

proving 1.

To show that $r_{k+1} \perp \text{span}\{r_0, \ldots r_{k-1}, r_k\}$, we first compute $\langle r_{k+1}, r_j\rangle$ for $j < k$:

$$\langle r_{k+1}, r_j\rangle = \langle r_k, r_j\rangle - \alpha_k \langle Ap_k, r_j\rangle = \langle r_k, r_j\rangle - \alpha_k \langle p_k, r_j\rangle_A.$$

The first inner product vanishes by property 2 of the IH. Since $r_j \in \mathcal{K}_{j+1}$ by property 1, and since the $\{p_0, \ldots p_j\}$ are a mutually A-orthogonal basis for $\mathcal{K}_{j+1}$ by property 4, we necessarily have that r_j is A-orthogonal to p_k. Hence $\langle r_{k+1}, r_j\rangle = 0$ for $j < k$.

Now consider the case $j = k$ case:

$$\langle r_{k+1}, r_k\rangle = \langle r_k, r_k\rangle - \alpha_k \langle p_k, r_k\rangle_A$$

where

$$\alpha_k = \frac{\langle b, p_k\rangle}{\langle p_k, p_k\rangle_A}.$$

If we can show that also

$$\alpha_k = \frac{\langle r_k, r_k\rangle}{\langle p_k, r_k\rangle_A}, \tag{26}$$

then property 2 will result. First note that

$$p_k = r_k - \big[\text{linear combination of } \{p_0, \ldots, p_{k-1}\}\big], \tag{27}$$

so that by property 4 of the IH we conclude $\langle p_k, p_k\rangle_A = \langle p_k, r_k\rangle_A$. Next note that $b = r_k + Ax_k$ so that

$$\langle b, p_k\rangle = \langle r_k, p_k\rangle + \langle Ax_k, p_k\rangle = \langle r_k, p_k\rangle + \langle x_k, p_k\rangle_A .$$

Since $\mathcal{K}_k = \operatorname{span}\{p_0, \ldots, p_{k-1}\} = \operatorname{span}\{r_0, \ldots, r_{k-1}\}$, by (27) and property 2 of the IH we have $\langle r_k, p_k\rangle = \langle r_k, r_k\rangle$. Since x_k is defined to be the A-orthogonal projection of $x^\star$ onto $\mathcal{K} = \operatorname{span}\{p_0, \ldots, p_{k-1}\}$, by property 4, $\langle x_k, p_k\rangle_A = 0$. Hence (26) indeed holds, so that property 2 is established.

To show property 3, note that we just showed $r_{k+1} \perp \mathcal{K}_{k+1}$. So, let $w \in \mathcal{K}_k$ so that $Aw \in \mathcal{K}_{k+1}$. Then

$$\langle r_{k+1}, Aw\rangle = \langle r_{k+1}, w\rangle_A = 0$$

and hence $r_{k+1} \perp_A \mathcal{K}_k$, proving 3.

Finally, we want to show $p_{k+1} \perp_A \operatorname{span}\{p_0, \ldots, p_k\}$. First, let $j < k$ and consider $\langle p_{k+1}, p_j\rangle_A$. By definition, p_{k+1} is the projection of r_{k+1} A-orthogonal to $\operatorname{span}\{p_0, \ldots, p_{k-1}, p_k\}$. From property 3, however, we know that r_{k+1} is A-orthogonal to $\mathcal{K}_k$; hence the projection simply reduces to

$$p_{k+1} = r_{k+1} - \frac{\langle r_{k+1}, p_k\rangle_A}{\langle p_k, p_k\rangle_A}.$$

By the IH, $\langle p_k, p_j\rangle_A = 0$ and thus we simply have

$$\langle p_{k+1}, p_j\rangle_A = \langle r_{k+1}, p_j\rangle_A .$$

This also vanishes, however, since $p_j \in \mathcal{K}_{j+1}$ and $r_{k+1} \perp_A \mathcal{K}_k \supseteq \mathcal{K}_{j+1}$ for $j < k$. Hence $p_{k+1} \perp_A \operatorname{span}\{p_0, \ldots, p_{k-1}\}$, and it remains to show $\langle p_{k+1}, p_k\rangle_A = 0$. By inspection

$$\langle p_{k+1}, p_k\rangle_A = \langle r_{k+1}, p_k\rangle_A - \frac{\langle r_{k+1}, p_k\rangle_A}{\langle p_k, p_k\rangle_A}\langle p_k, p_k\rangle_A = 0$$

as desired. □

As a brief corollary, we note from (22) that $x_k = x_k + \alpha_k p_k$ where

$$\alpha_k = \frac{\langle b, p_k\rangle}{\langle p_k, p_k\rangle_A}.$$

In the course of proving property 2 in the theorem above, we showed $\langle b, p_k\rangle = \langle r_k, r_k\rangle$, and hence:

Corollary 6.5.

$$\alpha_k = \frac{\langle r_k, r_k\rangle}{\langle p_k, p_k\rangle_A}. \tag{28}$$

We also know that $p_k = r_k + \beta_{k-1}p_{k-1}$ where

$$\beta_{k-1} = \frac{\langle r_k, p_{k-1}\rangle_A}{\langle p_{k-1}, p_{k-1}\rangle_A}.$$

It can be also be shown from the theorem that:

Exercise 6.6.

$$\beta_{k-1} = \frac{\langle r_k, r_k\rangle}{\langle r_{k-1}, r_{k-1}\rangle}. \tag{29}$$

We note both (28) and (29) are used in practical implementations of CG.

6.2. **Relationship to polynomial approximation.** In the previous subsection we characterized the CG method as producing iterative approximations to the minimizer x^* of (19) using

$$x_k = \operatorname*{arg\,min}_{x\in\mathcal{K}_k} \|x^* - x\|_A.$$

If we define the error $e_k := x^* - x_k$, then equivalently we seek to minimize $\|e_k\|_A$ at each step of the iteration. Since $x^* = A^{-1}b$ and $x_k \in \mathcal{K}_k$, we have

$$\begin{aligned} e_k &= A^{-1}b + c_1 b + c_2 Ab + \ldots + c_k A^{k-1}b \\ &= (I + c_1 A + c_2 A^2 + \ldots + c_k A^k)A^{-1}b \\ &=: q(A)A^{-1}b \end{aligned}$$

for some constants $c_1, c_2, \ldots, c_k$. For $x_0 = 0$, $e_0 = x^* = A^{-1}b$ and hence CG can be framed as the approximation problem:

$$\text{Find } q \in \tilde{\mathbb{P}}^k \text{ such that } \|q(A)e_0\|_A \text{ is minimized.}$$

Here $\tilde{\mathbb{P}}^k = \{q \in \mathbb{P}^k : q(0) = 1\}$. If let $\sigma(A)$ denote the spectrum of the SPD matrix A, then we have the following result:

Theorem 6.7.

$$\frac{\|e_k\|_A}{\|e_0\|_A} = \inf_{q \in \tilde{\mathbb{P}}^k} \frac{\|q(A)e_0\|_A}{\|e_0\|_A} \leq \inf_{q \in \tilde{\mathbb{P}}^k} \max_{\lambda \in \sigma(A)} |q(\lambda)| \tag{30}$$

Proof. Since the first equality in (30) has already been shown, it remains to derive the upper bound. First we compute $\|e_0\|_A$; since A is SPD, there is an orthonormal (with respect to the standard l^2 inner product) basis of eigenvectors $\{v_j\}_{j=1}^n$, so that $e_0 = \sum_{j=1}^n \gamma_j v_j$ for some scalars $\gamma_1, \ldots, \gamma_n$. Hence

$$\|e_0\|_A^2 = \left\langle \sum_{j=1}^n \gamma_j v_j, A \sum_{j=1}^n \gamma_j v_j \right\rangle = \sum_{j=1}^n \gamma_j^2 \lambda_j$$

where λ_j is the eigenvalue corresponding to v_j. Since $A^l v_j = \lambda_j^l v_j$ for any natural number l, we know $q(A)v_j = q(\lambda_j)v_j$ for any $q \in \tilde{\mathbb{P}}^k$. It then follows that

$$\|q(A)e_0\|_A^2 = \sum_{j=1}^n \gamma_j^2 \lambda_j (q(\lambda_j))^2$$

and thus

$$\frac{\|e_k\|_A^2}{\|e_0\|_A^2} = \inf_{q \in \tilde{\mathbb{P}}^k} \frac{\sum_{j=1}^n \gamma_j^2 \lambda_j (q(\lambda_j))^2}{\sum_{j=1}^n \gamma_j^2 \lambda_j} \leq \inf_{q \in \tilde{\mathbb{P}}^k} \max_{\lambda \in \sigma(A)} (q(\lambda))^2.$$

Taking the square root of both sides of the equality gives the result. $\square$

Corollary 6.8. *If A has μ distinct eigenvalues, then CG will converge in at most μ iterations:*

$$\|e_\mu\|_A = 0.$$

Proof. Let $\lambda_1, \lambda_2, \ldots, \lambda_\mu$ be the distinct eigenvalues, and define

$$p(z) = C(z - \lambda_1)(z - \lambda_2) \ldots (z - \lambda_\mu)$$

where C is chosen such that $p(0) = 1 \implies p \in \tilde{\mathbb{P}}^\mu$. By construction of course, $p(\lambda) = 0$ for any $\lambda \in \sigma(A)$, and hence $\|e_\mu\|_A / \|e_0\|_A \leq 0$, i.e. CG has converged by the μ-th iteration. $\square$

References

[1] N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM, 1997.